Response to Reviewers

Xiaoxu Shi[1*], Jiping Liu[2,1*], Chaoyuan Yang[1] and Gerrit Lohmann[3]

[1]Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), Zhuhai, China.
[2]School of Atmospheric Sciences, Sun Yat-sen University, Zhuhai, China.
[3]Alfred Wegener Institute, Helmholtz Center for Polar and Marine Research, Bremerhaven, Germany.

*Corresponding author(s). E-mail(s): Xiaoxu Shi (shixiaoxu@sml-zhuhai.cn); Jiping Liu (liujp63@mail.sysu.edu.cn);

Response to Reviewers

Manuscript: Glacial–interglacial regime shift in Southern Ocean dense water formation Authors: Xiaoxu Shi, Jiping Liu, Chaoyuan Yang, Gerrit Lohmann

We thank the editor and the four reviewers for the care they have taken with this manuscript. The reports are detailed and constructive, and several of them identify a problem that we had handled too lightly in the submitted version, namely the disagreement between our pre-industrial result and observation-based transformation estimates for the modern Southern Ocean. We took that criticism seriously and spent most of the revision on it. Rather than defend the original one-sentence explanation, we ran the diagnostic tests that Reviewers 2 and 3 proposed. The outcome is more informative than we expected, and it has changed how we frame the paper.

In brief, the disagreement has two separable causes. The first is the integration domain, as 42.7% of the area we integrated over lies outside the winter sea ice edge and is therefore water that Pellichero et al. did not include. The second is the density range. Resolving the transformation by density class over the sea ice zone alone shows that our pre-industrial state is not uniformly thermally driven. Only one class is thermally dominated, the band around σ_2 36.75 to 37.00 where heat contributes 70% of the transformation. Every denser class is haline dominated, with the sea ice term supplying 93% of the transformation above σ_2 37.00, and the lightest classes are dominated by the remaining freshwater terms. The thermal signature that drew the reviewers' attention is therefore confined to a narrow band at intermediate and mode water densities, and it sits lighter than the bottom water cell that Pellichero et al. describe.

The glacial states leave no such ambiguity. Every density class in the Last Glacial Maximum and Marine Isotope Stage 3 is haline dominated, with thermal shares between 4% and 28% and sea ice alone accounting for most of the transformation in the dense classes. The shift from a partly thermal interglacial to an unambiguously haline glacial is visible class by class rather than only in the integrated totals, which is the result the paper rests on.

A residual discrepancy remains after both effects are accounted for. Our pre-industrial Southern Ocean is too warm and too fresh near Antarctica, a bias shared by most coupled models, and our dense water forms by open-ocean convection in the Weddell gyre rather than by dense shelf water overflow. Both push the transformation toward lighter and more exposed water than the observations sample. We now state this plainly and quantify it. What the paper claims, however, is the relative change between climate states rather than the absolute densities, and a bias that is present in every simulation cannot generate the glacial to interglacial contrast between them. On that basis we consider the regime shift robust, and we have bounded the rest of our claims accordingly.

directly, ideal age throughout the global ocean, and a fuller treatment of the Southern Annular Mode; discussed the study by Chen et al. (2025) that Reviewer 2 noted; and worked through every specific comment. Five new figures have been added, along with a boundary condition table and a Source Data file.

For convenience the principal new figures are reproduced in this letter, so that the reviewers can assess them conveniently.

Reviewer comments are reproduced in full below in black, each followed by our response in blue; text quoted from the revised manuscript is given in blue italics. Line numbers in our responses refer to the revised manuscript unless stated otherwise. All changes are marked in the accompanying tracked-changes file.

Draft covering Reviewer 1 only. The replies to Reviewers 2 and 3 are unchanged from the previous version and are not reproduced here, pending Jiping's review.

Response to comments by Reviewer #1

The manuscript presents a comprehensive investigation of Southern Ocean dense water formation across a range of glacial and interglacial climates using the AWI-ESM2 Earth system model. By applying a water mass transformation (WMT) framework, the authors successfully separate the relative roles of thermal and haline buoyancy forcing and demonstrate a compelling regime shift between interglacial and glacial climates. The manuscript is well structured, the analyses are technically rigorous, and the decomposition of the different buoyancy contributions provides valuable physical insight. The use of WMT diagnostics allows the underlying mechanisms to be quantified rather than inferred from changes in circulation alone. Overall, I found the study scientifically interesting and suitable for publication after major revisions. My comments below are primarily intended to strengthen the robustness of the conclusions and improve the clarity of the presentation.

We thank the reviewer for the constructive general comments and a thorough list of specific comments.

General comments

Model evaluation and confidence in simulated dense water formation

The conclusions of this study rely entirely on a single Earth system model. While I do not consider this to be a major limitation by itself, I think the manuscript would benefit from a more thorough discussion of how well AWI-ESM2 represents present-day Southern Ocean dense water formation and Antarctic Bottom Water formation. The manuscript briefly states that the PI simulation produces comparable WMT magnitudes to observational estimates, but Figure A4 only shows the model results, making it difficult for the reader to evaluate this statement. Given the importance of the model for all subsequent paleoclimate interpretations, I encourage the authors to provide stronger evidence supporting its performance. This could include a direct comparison with observational WMT estimates, or a more detailed discussion of known model strengths and biases in Southern Ocean stratification, sea ice, mixed-layer depth, and Antarctic Bottom Water formation. Even if such comparisons already exist in previous AWI-ESM2 publications, briefly summarising them here would substantially increase confidence in the conclusions.

Authors' response:
We agree, and we have restricted this reply to the pre-industrial simulation, which is what the reviewer asks about. The observational comparison comes first, then the model biases.
The reviewer is right that Figure A4 showed only model curves. We now place our pre-industrial transformation beside the observation-based estimate of Pellichero et al. (2018), computed over the same domain and the same season as theirs.

Pellichero et al. (2018) derived an observation-based estimate of the overturning sustained by surface buoyancy fluxes in the Southern Ocean sea ice sector. We compare against them because their sector, the region enclosed by the September 15 per cent sea ice contour, is the part of our domain where the comparison is meaningful, and because theirs is the estimate most often set against model results of this kind. To match them we recomputed our transformation over that sector, reproducing their domain to within 2.5 per cent in area, and took the annual mean rather than the austral winter mean used elsewhere in the paper.

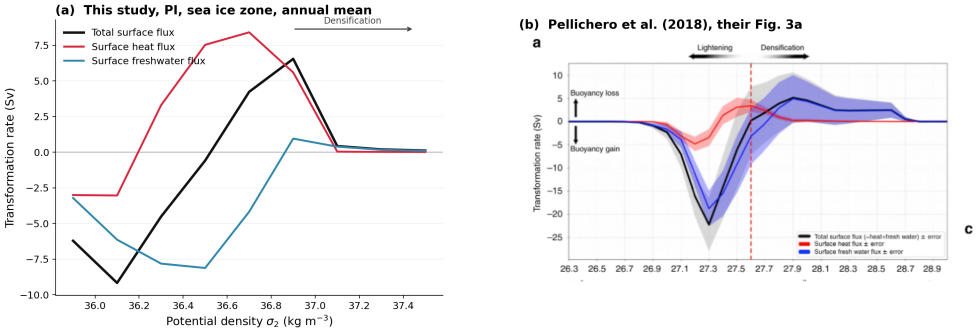


Figure R1.1 (new analysis). (a) Simulated pre-industrial transformation in the seasonal sea ice zone, annual mean, decomposed into the heat and freshwater contributions. (b) The observational estimate of Pellichero et al. (2018), their Fig. 3a, reproduced for comparison. The density axes differ deliberately: ours is potential density σ_2, theirs neutral density γ. Our simulated surface water is lighter than observed, so plotting both on one axis would imply a point-by-point correspondence that does not exist.

The two curves share their structure. Both show buoyancy gain at the lighter classes, a zero crossing, and densification at the heavier classes, with the freshwater term carrying the light end and both curves converging towards zero at the dense end. In our pre-industrial state the transformation reaches a minimum of -9.2 Sv, crosses zero, and peaks at +6.6 Sv; Pellichero et al. report a minimum near -22 Sv and a densification peak of 5 plus or minus 5 Sv. The sign structure and the order of magnitude of the densification branch therefore agree, which is the comparison the reviewer asks for.

Two differences are real and we state them rather than smoothing them over. Our light-end buoyancy gain is weaker than theirs, and our whole curve sits at lighter densities, because the simulated Southern Ocean surface is fresher than observed. Both follow from the same bias, which we quantify under model limitations below.

The reviewer, and Reviewer 3, also asked whether we could compute an observational transformation ourselves from a reanalysis rather than citing a published one. We examined this and it is not feasible, for three reasons.

First, the data do not exist. A transformation budget needs the net surface heat flux, the net surface freshwater flux, and the sea ice thermodynamic salt flux. We checked the Copernicus ocean reanalyses, namely GLORYS12, the GREP ensemble and the analysis-forecast product, which carry temperature, salinity, velocity and sea ice state but no surface flux fields at all. ORAS5 does carry a net downward heat flux and a net upward water flux, but we tested its water flux inside the September ice zone and found no freezing signal: at peak freezing it is -1.2e-5 kg m-2 s-1 against -1.6e-5 at peak melt, so the field is evaporation minus precipitation minus runoff and excludes the sea ice term. The sea ice contribution is precisely the term that decides the question at issue, and no reanalysis we could find supplies it.

Second, Pellichero et al. themselves reject that approach for this region. They write that ocean buoyancy fluxes are commonly obtained from atmospheric reanalysis datasets, but that *"in the Southern Ocean such reanalysis products are poorly constrained, and essentially unusable in the seasonally sea-ice covered region."* Rather than use one, they solve the mixed layer heat and salt budgets for the surface fluxes as the unknown, using the observed seasonal cycle of mixed layer temperature, salinity and depth together with Ekman and geostrophic advection and entrainment. The hydrography comes from Argo floats, ship-based observations and instrumented marine mammals, the only platform that samples under winter sea ice; the ice edge comes from SMMR and SSM/I passive microwave data for 2000-2015, and the geostrophic term from AVISO mean dynamic topography for 1993-2012. We cannot reproduce that calculation, because it rests on a seal-tag mixed layer database those authors assembled and we do not have. Computing a reanalysis transformation anyway would mean doing the very thing their paper argues cannot be done in this region.

Third, the two quantities are not identical even in principle. What they solve for is the surface flux together with the diapycnal mixing term, because the mixed layer budget cannot separate the two. Our model diagnoses the surface flux alone. Some discrepancy is therefore expected on definitional grounds, before any model bias is considered, and the agreement in shape shown in Figure R1.1 should be read in that light.

We have therefore compared against their published result rather than recomputing one, and we say so in the Discussion.

A separate point is that the published evaluations of this model configuration do not examine the Southern Ocean at all. We checked, and none of them assesses Southern Ocean stratification, mixed layer depth, sea ice or overturning against observations. What they do cover we now summarise in the Methods, with the agreements and the disagreements given equally. In the revised manuscript the added passage reads:

"Because the balance of agreement and disagreement in those evaluations bears on how far the present results can be trusted, we summarise it here rather than leaving it to be sought in the earlier papers. For the

pre-industrial state, Shi et al. (2025) find precipitation in reasonable agreement with the GPCP observations, with local wet biases at the position of the Intertropical Convergence Zone in both solstitial seasons, a bias common to other coupled models. For the interglacials, Shi et al. (2022) compare both generations of AWI-ESM against the pollen compilation of Bartlein et al. and the multi-archive compilation of Turney and Jones. The model reproduces the mid-Holocene warming over western Europe and the north-south dipole of the temperature anomaly over eastern North America, and it captures the sign and magnitude of the wetting over the southern Sahara and South Asia. It disagrees with the proxies over North Africa, where the simulated mid-Holocene is cooler than the pre-industrial while the reconstructions show warming, and it underestimates the mid-Holocene wetting over the northern Sahara. For the last interglacial the simulated Northern Hemisphere warming and the tropical and North African cooling both match the records, but the Antarctic warming that the proxies show is absent, which Shi et al. (2022) attribute to the PMIP4 protocol holding the last interglacial ice sheet orography fixed at its pre-industrial configuration. For the glacial states, Shi et al. (2023) document the Last Glacial Maximum configuration against the Bartlein pollen compilation and MARGO sea surface temperatures, and Shi et al. (2025) find that model and proxies agree on the pronounced drying over most Northern Hemisphere continents at the Last Glacial Maximum, while for Marine Isotope Stage 3 no comprehensive precipitation compilation exists and the comparison rests on individual sites. Across all four paleoclimate states the simulated anomalies are consistent with the PMIP4 ensemble. We note that none of these evaluations examines Southern Ocean hydrography, sea ice or overturning against observations, so they constrain the large-scale climate response of the model rather than the specific fields on which the present analysis depends."

Because the Southern Ocean is left unevaluated in that literature, the diagnosis of open-ocean convection reported here is our own, from our own output, and we label it as such rather than attributing it to a published evaluation.

Boundary conditions

The description of the prescribed boundary conditions for the paleoclimate simulations could be expanded. In particular, the glacial simulations use the GLAC-1D ice-sheet reconstruction, but it is not entirely clear how the ice-sheet changes are implemented in the model (e.g., topography, land-sea mask, ice-sheet extent, bathymetry, or other boundary conditions). As Antarctic dense water formation is highly sensitive to such factors, providing additional details on the experimental setup would help readers assess the robustness of the simulations. This is not a request for additional simulations, but rather for a clearer discussion of the robustness of the conclusions and

the extent to which they may depend on the chosen model configuration and forcing datasets. A summary table listing the prescribed greenhouse gas concentrations, orbital parameters, and ice-sheet boundary conditions for each experiment would also improve the readability of the Methods section.

Following the reviewer's suggestion, we have provided a summary table (Table R1, manuscript Table 1), which lists the greenhouse gas concentrations, orbital parameters and ice sheet configuration for all five experiments. The values are those actually used in the runs rather than the protocol defaults, and we reproduce the table here so the reviewer does not have to open the manuscript to check them.

	CO_2 (ppm)	CH_4 (ppb)	N_2O (ppb)	Eccentricity	Obliquity (°)	Perihelion (°)	Ice sheet / bathymetry
PI	284.3	808.2	273.0	1850 CE	1850 CE	1850 CE	modern
MH	264.4	597.0	262.0	0.018682	24.105	180.87	modern
LIG	275.0	685.0	255.0	0.039378	24.040	95.41	modern
LGM	190.0	375.0	200.0	0.018994	22.949	294.42	GLAC-1D 21 ka
MIS3	210.5	556.2	247.4	0.013676	23.2591	25.99	GLAC-1D 38 ka

Table R1. Boundary conditions prescribed in the five equilibrium simulations. Orbital parameters for PI are those of 1850 CE. Greenhouse gas concentrations are volume mixing ratios.

We have also expanded the text describing the boundary conditions used in the experimental design, especially the implementation of the GLAC-1D ice-sheet reconstruction. Now reads:

"We perform five equilibrium simulations spanning distinct climate states including three interglacial periods, i.e., pre-industrial (PI), mid-Holocene (MH), and last interglacial (LIG), and two glacial periods, i.e., the Last Glacial Maximum (LGM) and Marine Isotope Stage 3 (MIS3). The boundary conditions are configured following the criteria of PMIP4 and are summarised in Table 1. Orbital parameters are calculated following , and the greenhouse gas concentrations are taken from multi-archive reconstructions from ice core records and recent measurements of firn air and atmospheric samples . For LGM and MIS3, we fix the boundary conditions at 21 ka and 38 ka respectively, with the topography and ice-sheet properties deriving from the GLAC1D reconstruction . The GLAC-1D reconstruction enters the model through several boundary fields at once, and it is clearest to list them by the component that uses them. Specifically, the atmosphere receives the ice sheet extent and surface elevation as orography, together with the land-sea mask implied by the lowered sea level, so that the glacial topography and coastline are felt by the circulation. The land surface receives the same mask and elevation, and from them the river routing and the vegetation distribution, both of which are recomputed for the glacial geometry rather than held at their modern values. The ocean receives the bathymetry and the coastline. Because the ocean component uses an unstructured mesh, a changed coastline cannot

be imposed on the existing grid and a dedicated mesh must be generated for each glacial period; we use meshes constructed for 21 ka and 38 ka respectively, while PI, MH and LIG share the modern-geometry mesh. Global mean ocean salinity is raised in the glacial experiments to account for the water stored in the ice sheets. No prescribed ice sheet meltwater flux is applied in any of the equilibrium simulations, and the model configuration does not include ice shelf cavities, so basal melt beneath floating ice shelves is not represented."

Clarifying several interpretations

Most interpretations are convincing, although a few statements could be presented more carefully or supported more explicitly. For example, some descriptions of the role of the Southern Annular Mode appear stronger than the presented evidence suggests, and several discussions would benefit from more direct references to the relevant figure panels.

We have moderated the Southern Annular Mode claims. The most overstated passage was the suggestion that the simulated coupling implies a route to marine proxy reconstruction; we have replaced it with a more careful statement noting that the accumulation rates and chronological precision required are rarely available around the Antarctic margin, a point Reviewer 3 also raised. We have also removed the claim that the circumpolar picture holds uniformly and now state explicitly where it does not, in particular the Ross Sea. Panel-level figure references have been added throughout, as requested in this comment and in comment 21.
In the revised manuscript the moderated passage now reads:

"Whether this persistent coupling could be exploited for proxy reconstruction is less clear than we previously suggested. Recovering an interannual-to-decadal signal from marine sediments would require accumulation rates and chronological precision that are rarely available around the Antarctic margin, where sedimentation is slow and bioturbation mixes the record over intervals far longer than the variability of interest. We therefore present the persistence of the coupling as a property of the simulated system rather than as a practical route to a marine Southern Annular Mode reconstruction."

Model limitations

Model limitations: The manuscript convincingly demonstrates that sea ice drives the transition from a thermally dominated to a haline-dominated dense water formation regime. However, the discussion would benefit from a brief consideration of the known biases and limitations of AWI-ESM2, particularly regarding Southern Ocean sea ice, mixed-layer depths, and Antarctic Bottom Water formation. Discussing how these

model characteristics may influence the proposed mechanism would help readers assess the robustness and broader applicability of the conclusions.

Authors' response:
This comment asks specifically about bias, where the general comment above asked about evaluation, so we treat the two separately. We take the three quantities the reviewer names in turn, and in each case compare the pre-industrial simulation against observations rather than against the other climate states.
Before doing so, one limitation underlies all three and is best stated once. We diagnosed where the model actually forms its dense water. In the pre-industrial simulation the Weddell sector accounts for 70 per cent of the area with a winter mixed layer deeper than 400 m south of 55°S and 97 per cent of the area deeper than 600 m, centred near 68°S in the interior of the gyre rather than over the shelf. This is open-ocean convection, not the shelf overflow pathway that operates in the real ocean. It is a limitation shared across the current model generation rather than particular to AWI-ESM2: Heuzé (2021) finds that 28 of 35 CMIP6 models form deep and bottom water by open-ocean convection that is too deep, too frequent or too extensive, with no model showing convincing shelf export in the Weddell Sea. We also now state that this configuration has no ice shelf cavities, so basal melt beneath floating ice is absent by construction.
Mixed layer depth. This is the clearest bias and the one most relevant to our mechanism. The model produces its pre-industrial deep mixed layers in the interior of the Weddell gyre near 68 deg S rather than over the continental shelf, so the convection is of the open-ocean type. The published evaluation of this model configuration does not itself examine Southern Ocean mixed layer depth, so this is our own diagnosis from our own output, and we state it as such. We therefore compare the simulated field directly against the ORAS5 ocean reanalysis, which does carry a mixed layer diagnostic for this region.
The broad pattern is reproduced. Both the model and the reanalysis keep shallow mixed layers over most of the open Southern Ocean, deepen towards the Antarctic margin, and place the deepest water of the sector in the Atlantic sector. Where they part company is the Weddell gyre interior: the model opens a patch deeper than 400 m that the reanalysis does not have at all, and its Weddell sector mean is 128 m against 71 m. This is the open-ocean convection described above, and the comparison localises it rather than leaving it as an assertion.

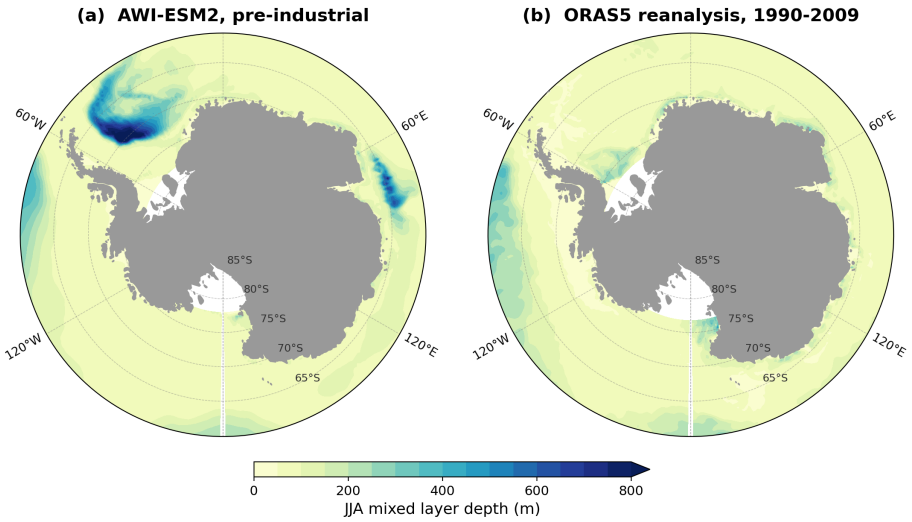


Figure R1.2 (new analysis). Winter mixed layer depth south of 60°S. (a) The pre-industrial simulation. (b) The ORAS5 ocean reanalysis, JJA climatology over 1990-2009. The large-scale pattern agrees: both are shallow over most of the open Southern Ocean, both deepen towards the Antarctic margin, and both show the deepest water of the sector in the Atlantic sector. The difference is where it matters for this study. The model places a large patch deeper than 400 m in the interior of the Weddell gyre, covering 1.9 per cent of the area south of 60°S against 0.0 per cent in the reanalysis, and its Weddell sector mean is 128 m against 71 m. Area-weighted over the whole sector the model gives 100 m against 78 m, so the bias is concentrated in the convecting patch rather than spread evenly. We note that the two use different mixed layer definitions, a 0.01 kg m^{-3} density threshold in ORAS5 and a bulk Richardson number criterion in the model, so the point-by-point difference is not purely model error.

Sea ice. The published evaluation reports that Antarctic sea ice in this configuration is too thin, hardly reaching 0.25 m in the central Weddell Sea in September, and that the Southern Ocean carries the model's most pronounced warm bias, with a sea surface temperature root-mean-square error of 1.43 K against the PHC climatology. A warm, thin-ice Southern Ocean is precisely the state in which open-ocean convection is favoured over shelf processes, so the documented biases and the convection behaviour we diagnose are consistent with one another.

Bottom water formation. The abyssal overturning cell of this configuration is reported at about 10 Sv, comparable to our pre-industrial value, but it is produced by the wrong pathway, as described above. We therefore present our results as a statement about how the partitioning of surface buoyancy forcing responds to changing ice cover, and explicitly not as a quantitative reconstruction of bottom water formation rates.

Taken together these are systematic biases: the mixed layers are too deep in the open ocean and too shallow on the shelf, the sea ice is too thin, and

the surface is too warm and too fresh, in every one of our simulations alike. That matters for how the results should be read. Because the same bias is present in all five experiments, it largely cancels in the differences between them, which is what the paper actually reports. We make no claim about the absolute transformation rate or the absolute density at which it occurs; we claim that the balance between the thermal and haline pathways shifts between climate states, and a bias common to every state cannot generate that shift.
The corresponding passage added to the Discussion reads:

"Independent evaluations of this model provide context for these limitations, and we summarise the relevant points here rather than leaving them to be sought in the model description literature. Sidorenko et al. (2019) evaluate the coupled configuration against observations and report that the abyssal overturning cell associated with Antarctic Bottom Water is reproduced with a maximum of about 10 Sv, comparable to the pre-industrial value obtained here. The same evaluation identifies the Southern Ocean as the region of the most pronounced warm bias, with a sea surface temperature root-mean-square error of 1.43 K against the PHC climatology at the resolution used here, and notes that Antarctic sea ice is too thin, hardly reaching 0.25 m in the central Weddell Sea in September. Their overall assessment is that although biases in the representation of water mass properties and ventilation mechanisms are present, the model still produces a reasonable density distribution that maintains realistic transports. A warm and thin-ice Southern Ocean is precisely the state in which open-ocean convection is favoured over shelf processes, so these documented biases are consistent with the convection behaviour we diagnose above, although we note that the published evaluation does not itself examine mixed-layer depth, polynyas or the convection pathway, and the diagnosis of open-ocean convection reported here is our own."

Latitude and longitude labels, and subpanel references

Please consider adding latitude and longitude labels to all map figures where appropriate. In addition, referring to specific figure subpanels throughout the text, rather than only the full figure number, would make the discussion easier to follow.

Subpanel references have been added throughout the Results and Discussion. On the map labels: the polar stereographic panels now carry labelled latitude circles, and the ventilation age maps carry labelled meridians as well. On the fifteen-panel composite figures we label the latitude circles only, because a full graticule with meridian labels on panels of that size was illegible in trial versions and crowded the panel margins. We are happy to add complete labelling if the reviewer prefers, or to enlarge those figures to accommodate it. As a concrete example, the ventilation age maps now carry labelled meridians at 60°W, 0°, 60°E and 120°E as well as labelled

latitude circles, and the mixed layer and composite figures carry labelled latitude circles; Figure R1.3 in this letter shows the result. The same treatment has been applied to every map figure in the revised manuscript and supplement.

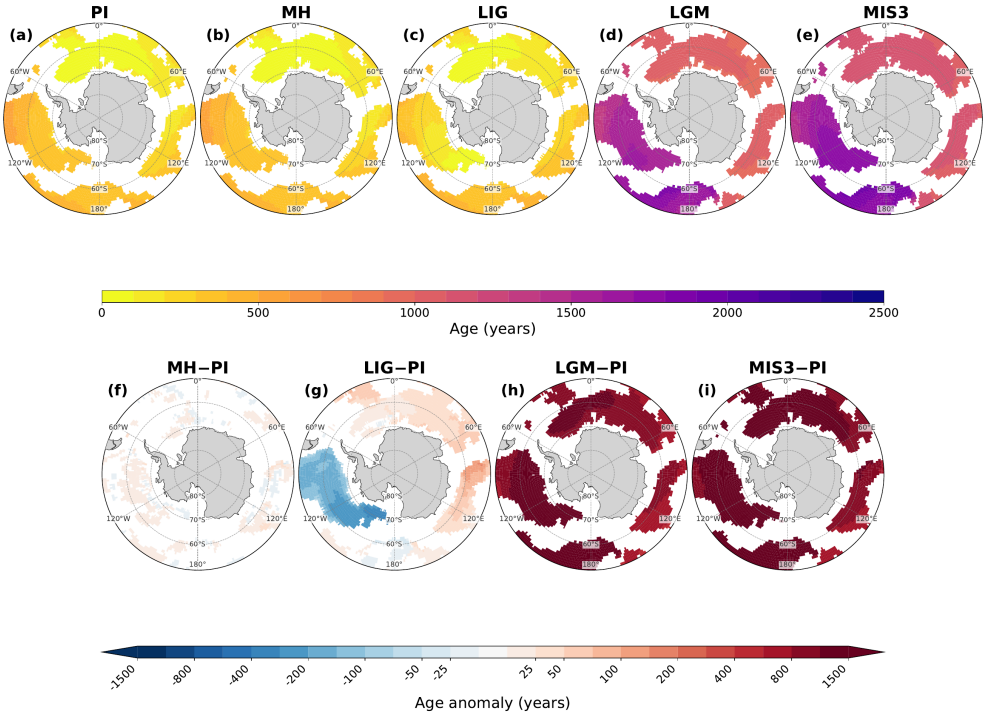


Figure R1.3 (revised manuscript figure). The ventilation age figure as an example of the labelling now applied throughout. Meridians are labelled upright inside the map boundary and latitude circles are labelled along the 180° meridian, at 60°S, 70°S and 80°S. Placing both inside avoids the labels of neighbouring panels running into one another, which happened when they were set outside the circular boundary, and it leaves more room for the maps themselves. The mixed layer, wind stress and composite figures have been treated the same way; please refer to the revised manuscript and supplement for the complete set.

Specific comments

1. LIG, LGM and MIS should be capitalised.

Thank you for catching this. The three abbreviations are now capitalised consistently throughout the manuscript, the figure captions and the supplement, and MIS is written as MIS3 everywhere so that it matches the column headings of the figures.

2. Line 21-22: The transition to SAM is quite abrupt.

The abstract now introduces the Southern Annular Mode through the mechanism that connects it to the mean state, noting that because the balance between the thermal and haline pathways is set by how much of the surface is ice covered, it also governs the response to internal atmospheric variability.
In the revised manuscript this now reads:

"The balance between these two pathways is set by sea ice extent, which is modulated by internal atmospheric variability. Further analysis suggests that the Southern Annular Mode modulates transformation by 3-8 Sv between its positive and negative phases in every climate state, working through heat loss when the ocean is open and through brine rejection when it is ice covered."

3. Line 24: "same mechanisms" sounds vague her without additional context. The abstract should be understandable on its own without requiring the reader to refer to the main text.

Replaced with an explicit statement of the two pathways, so the abstract no longer relies on the reader having seen the main text.
In the revised manuscript this now reads:

"The balance between these two pathways is set by sea ice extent, which is modulated by internal atmospheric variability. Further analysis suggests that the Southern Annular Mode modulates transformation by 3-8 Sv between its positive and negative phases in every climate state, working through heat loss when the ocean is open and through brine rejection when it is ice covered."

4. Lines 25-27: Similar to Comment 2 above, additional context is needed before introducing the deep ocean ventilation results. The motivation for analysing ventilation ages only becomes clear later in the manuscript, whereas the abstract should be self-contained.

The abstract now bridges from surface forcing to ventilation via the overturning circulation, stating that weaker glacial transformation coincides with a collapse of the simulated abyssal cell from about 10 to 2 Sv, before the ages are given. This also incorporates the new overturning analysis requested by Reviewer 3.
In the revised manuscript this now reads:

"Weaker glacial transformation coincides with a collapse of the simulated abyssal overturning cell from about 10 to 2 Sv and with older abyssal water. Ideal ages at 4000 m remain below 500 years during interglacials but exceed 750 years basin-wide during glacials, reaching 1500 years in the Pacific."

5. Lines 102-105: SAM is introduced only briefly here. Given its importance throughout the Results and Discussion, please consider providing a slightly more detailed introduction.

The introduction now defines the mode as the leading mode of extratropical Southern Hemisphere variability, describes what its positive phase does to the westerlies and hence to Ekman divergence, air-sea heat exchange and sea ice export, and cites the observational literature linking it to bottom water changes.
In the revised manuscript this now reads:

"The Southern Annular Mode is the leading mode of extratropical Southern Hemisphere circulation variability, describing a meridional redistribution of atmospheric mass between mid-latitudes and Antarctica . Its positive phase strengthens the circumpolar westerlies and displaces them poleward, which changes Ekman divergence, air-sea heat exchange, and the northward export of sea ice. In the present-day ocean these adjustments modulate mixed-layer depth and surface buoyancy forcing , and they have been linked to observed changes in bottom water properties . Whether the mode remains an effective modulator of dense water formation once the mean state changes, in particular once an expanded ice cover separates the ocean from the atmosphere, is not known."

6. Line 125: typo: 'more'.

Corrected ("moer" → "more").

7. Line 127: Figure numbering; A3 can't be called before A1 and A2.

The appendix figures have been reordered so that they are cited in numerical order.

8. Line 129: 'Here,'.

Corrected.

9. Line 140: As a result, the deep MLD signals, characteristic of open-ocean convection, largely.... Also, "vanish" is perhaps not the most scientific wording.

Rephrased to "are strongly reduced and become confined to narrow coastal bands and polynyas", and we now quantify the reduction rather than describing it qualitatively.
In the revised manuscript this now reads:

"The reduction is substantial rather than complete. Summed south of 55° S, the area with a winter mixed layer deeper than 400 m falls from 7.510ˆ11 mˆ2 in PI to 1.710ˆ11 mˆ2 in LGM and 1.610ˆ11 mˆ2 in MIS3, and the maximum mixed-layer depth decreases from 991 m to 819 m and 645 m. Mixed layers deeper than 600 m, which cover 2.410ˆ11 mˆ2 in PI, essentially disappear in both glacial states. The residual glacial signals in coastal regions are therefore genuinely shallower than their interglacial counterparts, consistent with dense water production that is confined to narrow coastal bands rather than distributed across the open gyres."

10. Line 141: These signals are not particularly visible even in the coastal regions (the darker blue shading in LGM is at ˜200m, so considerably shallower than the interglacials).

This is a fair objection and we have addressed it with numbers. Summed south of 55°S, the area with a winter mixed layer deeper than 400 m falls from 7.5×10^{11} m^2 in PI to 1.7×10^{11} m^2 in LGM and 1.6×10^{11} m^2 in MIS3, the maximum mixed-layer depth decreases from 991 m to 819 m and 645 m, and mixed layers deeper than 600 m essentially disappear in both glacial states. The text now says explicitly that the residual glacial signals are genuinely shallower than their interglacial counterparts. Figure R1.2 below shows the glacial fields on a scale that resolves them, and the reviewer's reading of the shading is correct and is now reflected in the text.
The reviewer's observation also pointed to a presentation problem, which we have fixed. On the shared 0–400 m scale of the original figure the glacial panels are almost featureless, because the 99th percentile of the glacial winter mixed layer south of 55°S is only 280 m (LGM) and 296 m (MIS3). Replotting those two states on their own 0–300 m scale makes the coastal cells visible.

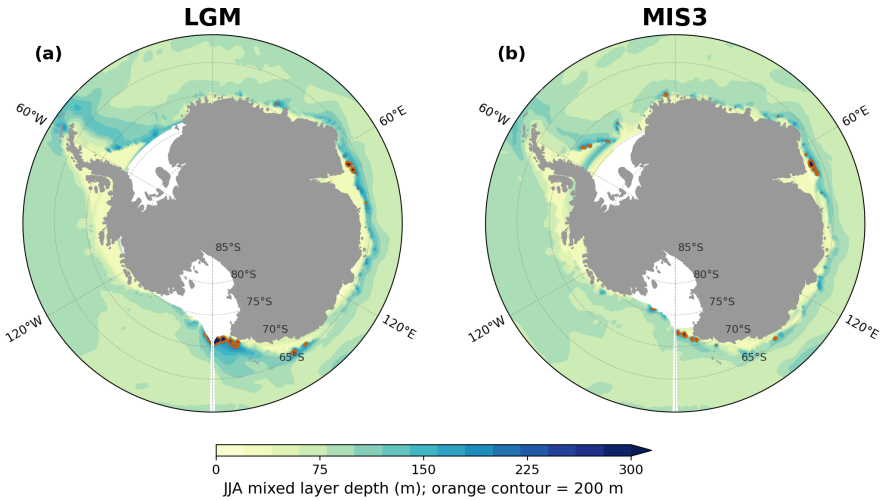


Figure R1.4 (new supplementary panel). (a) LGM and (b) MIS3 winter mixed layer depth south of 60°S, on a 0–300 m scale with the 200 m contour in orange. The tightened scale resolves discrete deep cells along the coast, in the Weddell, Prydz Bay and Ross sectors, which is where the glacial dense water is produced. This is the figure we now use in the supplement.

11. Line 134-141: Fig.1 i-l subpanels are not discussed at all.

A paragraph discussing the surface density panels has been added. It notes that the glacial density anomalies follow the salinity field rather than the temperature field, because the thermal expansion coefficient is small at

the near-freezing temperatures south of the ice edge, and points out the density-compensated LIG anomaly and the weak negative LGM anomalies in the Atlantic-Indian sector.
In the revised manuscript this now reads:

"The surface density anomalies (Fig. 1i-l) follow the salinity field rather than the temperature field in the glacial states. Cooling and salinification both act to densify the surface, but at the near-freezing temperatures that prevail south of the ice edge the thermal expansion coefficient is small, so the haline term sets both the pattern and the magnitude. In LIG the warm and salty Bellingshausen-Amundsen anomaly is close to density compensated (Fig. 1j), and the LGM retains weak negative density anomalies in parts of the Atlantic-Indian sector where surface freshening outweighs cooling."

12. Line 144-146: The LIG also exhibits similar behaviour, although the changes are smaller in some basins. Likewise, the LGM shows some negative anomalies in the Atlantic–Indian sector.

Both points are now stated in the text; see the response to comment 11.

13. Line 146-148: An explanation of why the wind stress changes do not lead to increased sea ice in the Ross Sea would be useful. Is this due to the presence of relatively warm water despite enhanced salinity, density, and Ekman transport in this region?

We checked this directly, and the answer turned out to be more interesting than the question assumed, so we set it out in full.
The mechanism the reviewer proposes is real, but it operates in the last interglacial rather than in the glacial states the comment refers to. Averaged between 60 and 75°S in winter, the glacial sea ice response is in fact fairly uniform around the continent: concentration rises by 0.62 in the Ross sector, 0.70 in the Weddell sector and 0.53 in the Adélie sector at the Last Glacial Maximum, and all three sectors cool to within about 0.1 K of the surface freezing point, so ice growth is not limited by surface heat content anywhere.

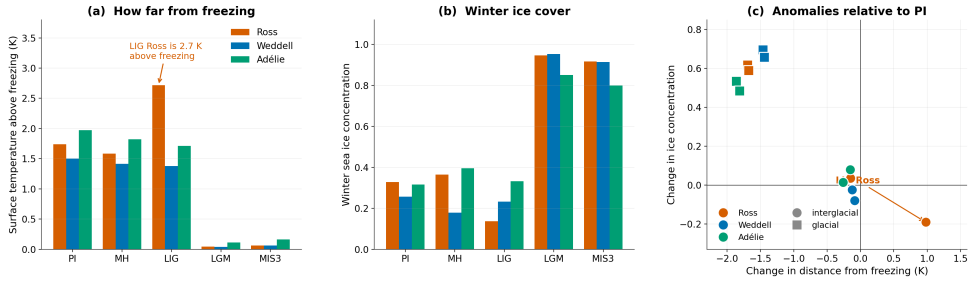


Figure R1.5 (new analysis). (a) Winter surface temperature above the local freezing point and (b) winter sea ice concentration, by sector and climate state. (c) The same quantities as

anomalies relative to PI. The Ross sector separates from the others only in the last interglacial, where it warms to 2.7 K above freezing and loses 0.19 of ice concentration, an order of magnitude more than the Weddell or Adélie response. In the glacial states all three sectors collapse onto the freezing point and gain ice comparably.

So in the last interglacial the Ross sector is 0.97 K warmer than in PI and sits 2.72 K above the freezing point, against changes below 0.3 K elsewhere, and its winter ice concentration falls by 0.19. There the wind can export ice efficiently but the water is too far from freezing for the exported ice to be replaced, which is precisely the reviewer's proposed mechanism. We have rewritten the passage to say this, rather than attributing the behaviour to the glacial wind response as the submitted version implied. This asymmetry is also why the Ross sector departs from the circumpolar pattern in the variability analysis.

14. Line 167: It would be helpful to quantify the thermal contribution, similar to the quantification provided for the haline contribution in line 157.

Added. The thermal term accounts for 70–85% of the total at the transformation maximum in the interglacials, against 15–30% for the haline term, and falls to 12–17% in the glacial states.
In the revised manuscript this now reads:

"This transformation is dominated by surface heat loss (red line), which accounts for 70-85% of the total at the transformation maximum, with limited sea ice extent restricting the haline contribution from brine rejection to a secondary role (15-30%; green line). Other freshwater fluxes (blue lines) oppose densification through net precipitation and river runoff. We note that this thermally dominated maximum occurs at densities that correspond to intermediate and mode waters rather than to bottom water. In the densest classes that the simulated surface fluxes reach ($\sigma_2 \geq 37.0$ kg m^{-3}), the PI transformation is already haline dominated, with the sea ice term contributing 0.79 Sv against 0.06 Sv from heat in the annual mean. The thermal dominance reported here is therefore a property of the integral over all density classes, and we quantify its sensitivity to the choice of integration domain and density range in the Discussion."

15. Fig. 2 Would it be possible to share a figure with the same y-axis limits for all panels? This may not be necessary for the final manuscript, but it would make it easier to compare the magnitude of the changes across the different climate states.

Yes. Supplementary Fig. S5 now shows the same transformation curves on common vertical and horizontal axes, so magnitudes and density positions can be compared directly across climate states and regions. We have kept the independently scaled version in the main text because the shape of the individual curves is otherwise hard to read, and the main text now points to the common-axis version explicitly.

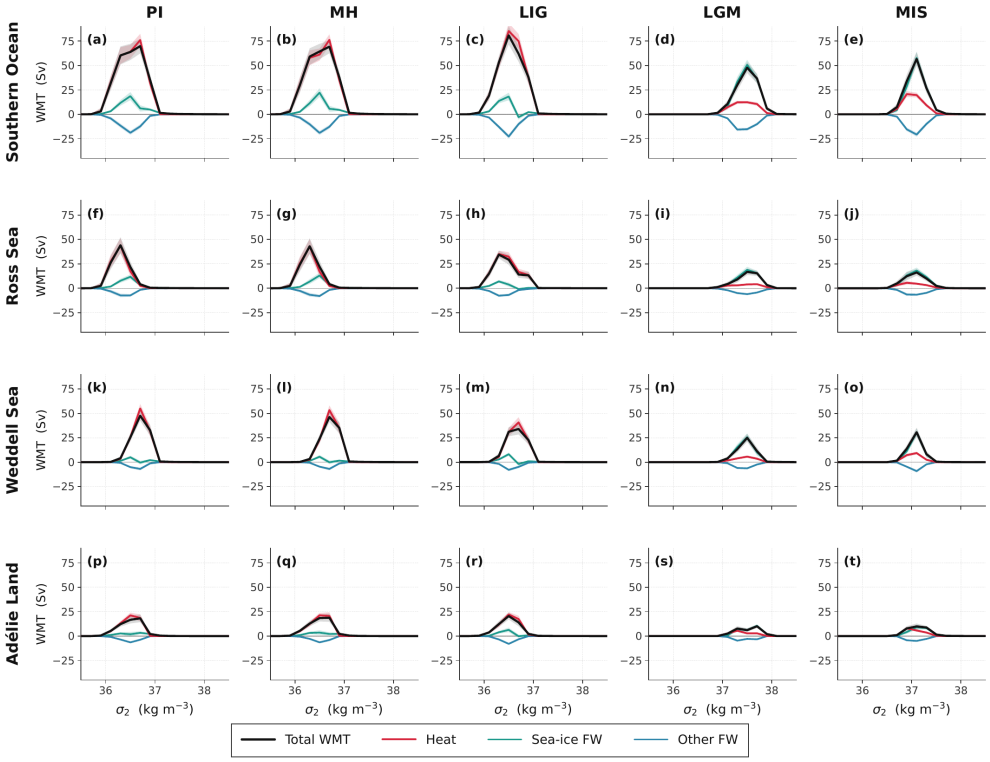


Figure R1.6 (new Supplementary Fig. S5). Winter transformation for the four sectors and five climate states on a common vertical and horizontal axis. Plotted this way the glacial weakening of the transformation maximum and its shift to denser classes are directly comparable between panels, which the independently scaled main-text version does not allow.

16. Line 174: "Section 4?"If this is intended to refer to the Methods section.

Corrected to refer to the Methods section by name.

17. Fig.3 Please consider adding the climate-state labels at the top of each column, as in the other figures.

Added, here and on the other multi-panel figures; Reviewer 3 made the same request.

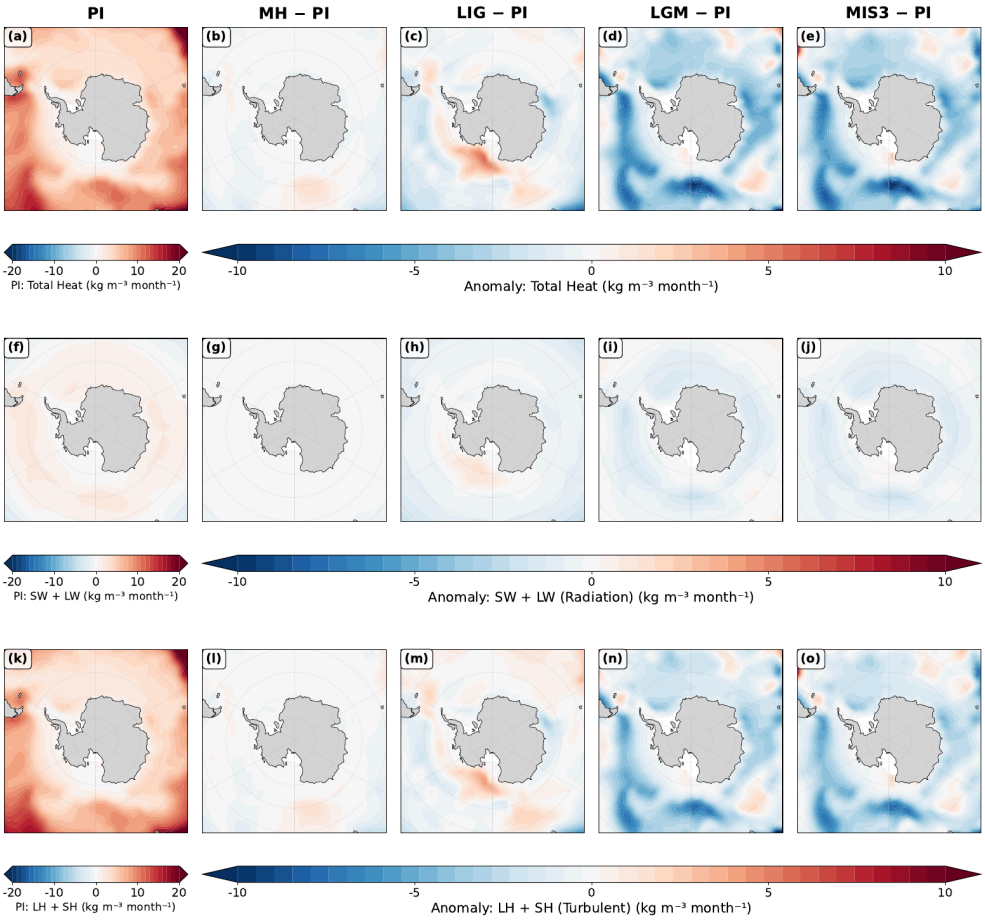


Figure R1.7 (revised manuscript Fig. 3). The surface density tendency figure with the climate-state headings added, so that the columns are identified in the same way as in the other figures. The freshwater counterpart, Fig. 4, has been given the same headings.

18. Line 180/181: The LIG is first described as "showing modest changes", but later as "exhibiting a pronounced increase". These descriptions appear inconsistent.

The two statements referred to different quantities, which was not clear. The changes are modest in the circumpolar integral and pronounced locally in the Ross Sea sector. The text now says so explicitly.

19. Could the authors briefly explain what is meant by thermal density tendency and sea-ice-driven density tendency in the methods.

Definitions have been added to the Methods, including the sign convention and the physical meaning of brine rejection versus melt.
In the revised manuscript this now reads:

"We refer throughout to the thermal density tendency, meaning the surface density change produced by the net heat flux acting through the thermal expansion coefficient, that is the -≤ft(/C_p)Q_net term, and to the sea-ice-driven density tendency, meaning the part of the haline term S F_fw that is carried by the thermodynamic growth and melt of sea ice. A positive value denotes densification of the surface water in both cases. Brine released during ice growth increases surface salinity and therefore density, whereas melting releases freshwater and reduces it."

20. Line 209: Please add the appropriate reference(s) here to support this statement.

We now cite Cerovečki et al. (2013) and Abernathey et al. (2016) in support of the statement that turbulent rather than radiative fluxes control the surface buoyancy budget.

21. Line216-246: Please refer to the relevant subpanels of Figure A5.

Done, and the same treatment has been applied throughout the Results and Discussion rather than only in the passage the reviewer names. As an example of the form these references now take:

"In PI, strong thermal densification (red shading) is concentrated in the open ocean (Fig. 3a), with maximal values reaching 5 10ˆ-6 kg m^{-3} s^{-1} in the circumpolar band between 50-65°S. This thermal densification is overwhelmingly controlled by turbulent fluxes (sensible and latent heat), with radiative fluxes playing a minor role in the buoyancy budget (Fig. 3f-j vs. Fig. 3k-o), consistent with observationally based surface buoyancy budgets for the Southern Ocean ."

22. Line 249, 255: The phrase "across at high latitudes" is somewhat unclear. Does this refer specifically to the Antarctic coastal region or the Southern Ocean south of a particular latitude? Please consider rephrasing for clarity.

Both instances have been rewritten. They now read "over the ice-free parts of the Southern Ocean south of about 50°S" and "south of the ice edge" respectively.

23. Could the authors clarify where the primary deep water formation regions occur in this model? Identifying these regions would also help assess whether the SAM-related changes shown in Figure 5 are directly influencing the main regions of dense water formation, as relatively little signal is apparent over the Weddell and Ross Seas.

We have added this diagnosis to the Results, and Figure R1.2 above shows it directly: the winter mixed layer depth for all five states. In the pre-industrial state the Weddell sector accounts for 70% of the area with winter mixed layers deeper than 400 m south of 55°S and 97% of the area deeper than 600 m, centred near 68°S in the gyre interior. The formation region is therefore the open Weddell gyre rather than the shelf, which is directly relevant to the model limitation discussed under model evaluation above. This also explains the pattern the reviewer noticed: the Southern Annular

Mode heat flux signal is strongest over open water, and in the glacial states the coastal regions where dense water is produced are precisely those insulated by ice, so the surface flux anomalies there are small even though the transformation response is not.

24. Lines 261-263: Please consider expanding on the physical mechanism linking positive SAM to freshwater loss and intensified brine rejection by additional explanation and supporting references.

Expanded, and we have added the two references the reviewer asks for. The passage now states the mechanism as a chain rather than asserting the endpoint, and cites the observational work on Antarctic Bottom Water production in the Cape Darnley polynya and on the sea ice contribution to Southern Ocean transformation.
In the revised manuscript this now reads:

"This pattern is controlled by sea ice processes, and the chain that produces it runs as follows. Strengthened and poleward-shifted westerlies increase the offshore component of the wind stress at the coast, which drives sea ice away from the continental margin. The divergence opens and widens coastal polynyas, exposing water already at the freezing point directly to a cold atmosphere. New ice therefore forms rapidly in those openings, and because ice rejects most of its salt as it grows, the salt is delivered to the surface ocean precisely where the water is coldest and densest. The ice exported offshore carries the compensating freshwater north, where it melts and freshens the surface. The result is the observed dipole, with salinification at the coast and freshening offshore, and it is the same mechanism identified in observations of Antarctic Bottom Water production in the Cape Darnley polynya and in the sea ice contribution to Southern Ocean water mass transformation more generally ."

25. Line 291, 294: Please check citation format.

Corrected.

26. Line 301:Figure A4 appears to show only model results. Please consider including the observational estimates in the figure (or referring to them directly) to support this statement. As noted above, it would also improve readability if the appendix figures were cited in numerical order.

We have added the quantitative comparison to the text and to the new Fig. 8, reproduced as Figure R1.1 above, which shows our transformation recomputed over the Pellichero domain alongside the standard domain. We report their published values directly in the Discussion, including the 5 ± 5 Sv transformation into denser classes and the stated factor of 2–5 by which the freshwater term exceeds the heat term in their sector, and we compare our numbers against them. We chose to make the comparison quantitatively in the text and in a dedicated figure rather than overplotting their curve, because their analysis is in neutral density over a different

density range and an overlay would misleadingly imply a point-by-point correspondence that does not exist; the density-range mismatch is itself one of our findings. The appendix figures are now also cited in numerical order. We note that this comment and the general comment on model evaluation above concern the same comparison, and we have kept the two answers consistent: the quantitative comparison with the published observational values is given once in the Discussion, and both replies point to the same new figure.

27. Lines 308-321: The discussion quantifies the interglacial changes (e.g., lines 315–316), but similar quantification is not provided for the glacial changes. For consistency, please consider including quantitative estimates for both.

Added. The glacial transformation maximum weakens to below 60 Sv and moves to σ_2 = 37.0–37.5 kg m^{-3}, with the thermal share falling to 12–17%. In the revised manuscript this now reads:

"The winter transformation maximum weakens to below 60 Sv and moves to σ_2 = 37.0-37.5 kg m^{-3}, and the thermal share of the transformation falls to 12-17% while the sea ice term becomes dominant. Extensive ice cover insulates the ocean from atmospheric heat exchange, and simultaneously intensifies haline forcing through enhanced brine rejection concentrated in coastal regions."

28. Lines 326 appear to repeat the statement made in line 323. In addition, the general statement in lines 323–325 does not appear to hold equally across all climate states. For example, the Ross Sea and Weddell Sea exhibit different behaviour for the peak-density classes.

The repetition has been removed by merging the two sentences. We have also added the qualification the reviewer asks for, noting that the Ross Sea departs from the circumpolar picture and that in the glacial states the signal is carried largely by the Weddell sector.
In the revised manuscript this now reads:

"This circumpolar picture does not hold uniformly in every sector. The Ross Sea in particular departs from it, showing weaker and at some densities opposing responses, and in the glacial states the circumpolar signal is carried largely by the Weddell sector while the Ross and Adelie sectors contribute little."

29. Line 339: Please provide a reference for the sediment-record evidence discussed here.

On reflection we think the passage was speculative rather than supported, and Reviewer 3 raised the same objection from the opposite direction, noting that accumulation rates around the Antarctic margin are very low. We have replaced the proposal with a statement of the difficulty, and now

present the persistence of the coupling as a property of the simulated system rather than as a practical route to a marine reconstruction.
In the revised manuscript this now reads:

"Whether this persistent coupling could be exploited for proxy reconstruction is less clear than we previously suggested. Recovering an interannual-to-decadal signal from marine sediments would require accumulation rates and chronological precision that are rarely available around the Antarctic margin, where sedimentation is slow and bioturbation mixes the record over intervals far longer than the variability of interest. We therefore present the persistence of the coupling as a property of the simulated system rather than as a practical route to a marine Southern Annular Mode reconstruction."

30. Section 4.1: Please define the full names of the abbreviated components.

Done, and the definitions are now given where the figures use the terms rather than only in the flux description.
In the revised manuscript this now reads:

"In the figures the radiative component denotes the sum of the shortwave and longwave fluxes, and the turbulent component denotes the sum of the latent and sensible heat fluxes."

In the revised manuscript this now reads:

"In the figures the radiative component denotes the sum of the shortwave and longwave fluxes, and the turbulent component denotes the sum of the latent and sensible heat fluxes."

31. Line 388: Please check the citation formatting.

Corrected.

32. Lines 394-403: Please consider including a summary table listing the prescribed boundary conditions for each experiment (e.g., greenhouse gas concentrations, orbital parameters, ice-sheet reconstruction, and any other relevant forcings). This would make the experimental setup considerably easier to follow.

Added as Table 1; see G2.